

Note 3.2 Subspaces

- **Definition of Subspace**

If S is a nonempty subset of a vector space V , and S satisfies the conditions

- $\alpha \mathbf{x} \in S$ whenever $x \in S$ for any scalar α .
- $\mathbf{x} + \mathbf{y} \in S$ whenever $\mathbf{x} \in S$ and $\mathbf{y} \in S$.

then S is said to be a **subspace** of V . And every subspace of a vector space is a vector space in its own right.

- **Remarks**

- Since $0 \cdot \mathbf{x} = \mathbf{0}$, $\mathbf{0} \in S$.
- We define the $\{\mathbf{0}\}$ as the **zero subspace** and the subspaces excluding zero subspace and V itself is **proper subspace**.
- To show a subset S of a vector space is a subspace, we need to prove that S is nonempty and that its closure properties in the definition. Since every subspace must contain the zero space, we can verify that S is nonempty by showing that $\mathbf{0} \in S$.

The Null Space of a Matrix

- **Definition**

Let A be an $m \times n$ matrix. Let $N(A)$ denote the set of all solutions of the homogeneous system $A\mathbf{x} = \mathbf{0}$. That is,

$$N(A) = \{\mathbf{x} \in \mathbb{R}^n | A\mathbf{x} = \mathbf{0}\}$$

and we call it the **null space** of A .

Proof of subspace:

Clearly $\mathbf{0} \in N(A)$ thus it is a nonempty set. For any $\mathbf{x} \in S$ and any scalar α ,

$A\alpha\mathbf{x} = \alpha A\mathbf{x} = \alpha\mathbf{0} = \mathbf{0}$ hence the first property satisfies. And for $\mathbf{x}, \mathbf{y} \in A$,

$A(\mathbf{x} + \mathbf{y}) = A\mathbf{x} + A\mathbf{y} = \mathbf{0} + \mathbf{0} = \mathbf{0}$ hence the second property satisfies and it is a subspace of $\mathbb{R}^n$.

The Span of a Set of Vectors

- **Definition**

Let $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ be the vectors in a vector space V . A sum of the form

$\alpha_1 \mathbf{v}_1 + \alpha_2 \mathbf{v}_2 + \dots + \alpha_n \mathbf{v}_n$ where α_k is scalars is called a **linear combination** of

$\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$. The set of all linear combinations of $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ is called the **span** of $\mathbf{v}_1, \dots, \mathbf{v}_n$, devoted by $\text{Span}(\mathbf{v}_1, \dots, \mathbf{v}_n)$.

- **Theorem 3.2.1**

If $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ are the elements of a vector space V , then $\text{Span}(\mathbf{v}_1, \dots, \mathbf{v}_n)$ is a subspace

of V .

Proof:

Clearly the span is a nonempty set. For the first property, suppose β is arbitrary scalar, then

$$\beta(\alpha_1 \mathbf{v}_1 + \alpha_2 \mathbf{v}_2 + \cdots + \alpha_n \mathbf{v}_n) = \alpha_1 \beta \mathbf{v}_1 + \alpha_2 \beta \mathbf{v}_2 + \cdots + \alpha_n \beta \mathbf{v}_n$$

is also the linear combination of those vectors. For the second property,

$$a_1 \mathbf{v}_1 + \cdots + a_n \mathbf{v}_n + b_1 \mathbf{v}_1 + \cdots + b_n \mathbf{v}_n = (a_1 + b_1) \mathbf{v}_1 + \cdots + (a_n + b_n) \mathbf{v}_n$$

is also the linear combination of those vectors.

- **Geometric Representation for $\mathbb{R}^3$**

In $\mathbb{R}^3$, the span of $\mathbf{e}_1$ and $\mathbf{e}_2$ represents the set of all vectors in 3-space in the x_1x_2 -plane.

And a vector $\mathbf{x}$ in $\mathbb{R}^3$ is in $\text{Span}(\mathbf{e}_1, \mathbf{e}_2)$ if and only if it lies in the x_1x_2 -plane in 3-space.

Similarly, given two vectors $\mathbf{x}, \mathbf{y}$ which are not collinear and not zero vector, then they can form a plane and it is the representation of $\text{Span}(\mathbf{x}, \mathbf{y})$.

Spanning Set for a Vector Space

- **Definition**

The set $\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ is a **spanning set** for V if and only if every vector in V can be written as a linear combination of $\mathbf{v}_1, \dots, \mathbf{v}_n$.

Exercise Section 3.2

- [Question 5th](#)

- Suppose $p(x) = x^4, q(x) = -x^4 + x^3$ (The even degree means the leading term has even degree), then their sum is not in the set so it is not subspace.
- Since zero vector is not in the set, the set is not the subspace of P_4 .
- The set is the polynomials whose constant is zero and it satisfies the closure property and nonempty. So it is the subspace.
- For $p(x), q(x)$, if their sum is a constant or their root is not the same, they will not have at least one real root. So it is not the subspace.

- [Question 18th](#)

Since $\mathbf{0} \in S$, if $S \neq \{\mathbf{0}\}$, then we suppose $\mathbf{a} \in S$. Since for every vector $\mathbf{s} \neq \mathbf{0}$, we have $\alpha \mathbf{a} = \frac{\mathbf{s}}{\alpha} \mathbf{a} = \mathbf{s}$. So $S = \{\mathbf{0}\}$ or $S = \mathbb{R}^1$.

- [Question 21st](#)

The element in S is the linear combination of $\mathbf{e}_1$ and in T is $\mathbf{e}_2$'s. Since for $\mathbf{x} \in S, \mathbf{y} \in T$, $\mathbf{x} + \mathbf{y} = \alpha \mathbf{e}_1 + \beta \mathbf{e}_2$, which is not in S or T . So it is not a subspace.

- [Question 23rd](#)

Actually, the two are not right in general. Note that the **plus** is not the same as the **disjunction**.